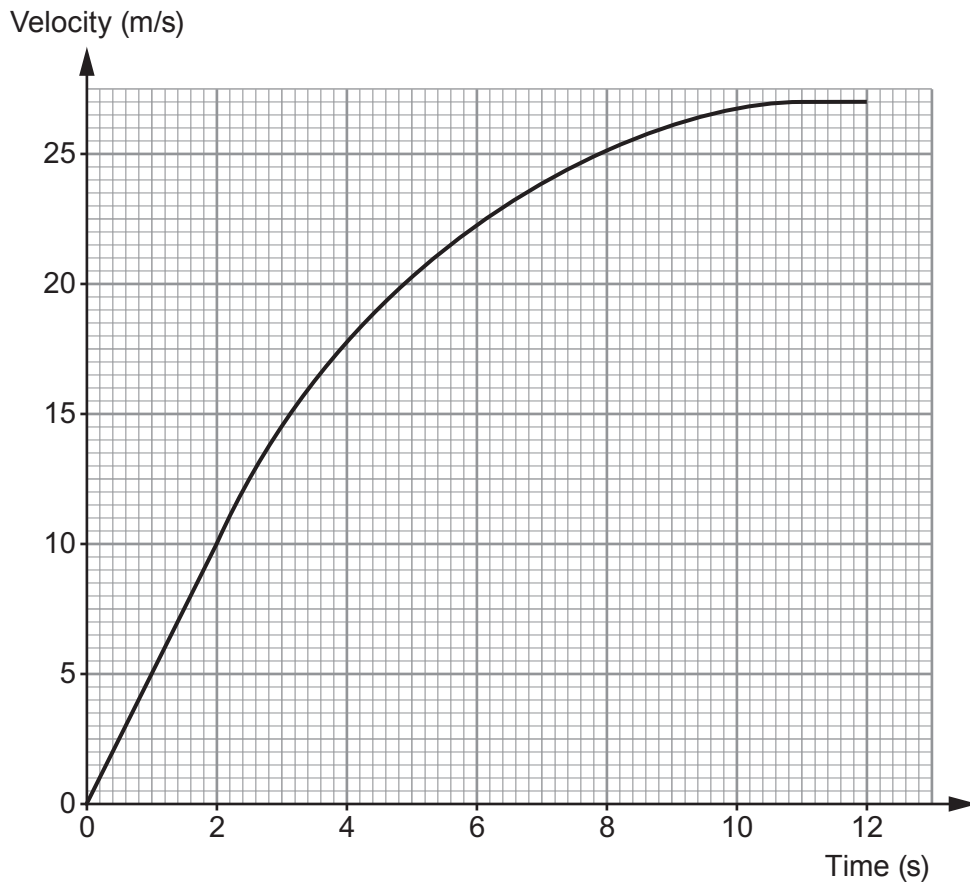


4. Manufacturers test new cars on a level track. In order to find out how long it takes them to accelerate to 27 m/s (60 mph), the cars are driven in a straight line at maximum power and the speed recorded. Data for one car is shown on the graph below.



- (a) Describe how the **acceleration** changes during the 12 s shown. [2]

.....

.....

.....

- (b) By drawing a suitable tangent and using:

acceleration = gradient of a velocity-time graph

calculate the acceleration of the car at 5 s. Give a unit with your answer. [3]

Acceleration = Unit =



- (c) Use an equation from page 2 to estimate the distance travelled by the car in the first 3 s. [2]

Distance = m

- (d) Another car of the same power and mass but with a more streamlined shape is tested. **Sketch on the grid below** the velocity-time graph for this car. [2]

